

Name: Ela Beren Yücel

Student ID: 34155

Proposal Title:

To what extent are people shifting from human-to-human interaction to human-to-AI interaction?

Artificial intelligence tools such as ChatGPT are becoming a common part of everyday life, and this may be changing how people communicate, ask for help, and interact with others. In the past, people usually relied on other people for many needs, such as asking questions, getting advice, studying together, joining forums, or even seeking emotional support. However, today some of these activities may be shifting toward human-to-AI interaction instead of human-to-human interaction. The aim of this project is to examine how this shift can be observed in real data and how strong this change is.

The main data source of this project will be Google Trends, which provides anonymized and aggregated data about how often certain terms are searched over time. I plan to collect data for keywords related to AI interaction (such as “ChatGPT”, “ask AI”, “AI girlfriend”, and “talk to AI”) and compare them with keywords related to human interaction (such as “forum help”, “relationship advice forum”, “study group”, and “AskReddit”). The data will be collected using Python with the pytrends library. It will mainly include weekly data from multiple years, resulting in several hundred observations for each keyword. Google Trends data is based on a sample of real search activity that is anonymized and aggregated, which makes it suitable as a behavioral proxy.

To make the analysis stronger, I will also use additional datasets. For example, I may use the 2025 Stack Overflow Developer Survey, which includes approximately 49,000 responses from 177 countries and provides detailed information about how people use AI tools and how they learn or solve problems. For emotional and personal uses of AI, I may use Pew Research survey data, which typically includes around 1,000 to 1,500 respondents in nationally representative samples. In addition, I may explore public health datasets such as NHIS, which includes approximately 30,000 to 35,000 individuals per year and provides variables related to counseling or therapy use.

In the later stages of the project, I also plan to apply machine learning methods to the cleaned datasets. First, I will create a classification task to examine whether AI-related search observations can be distinguished from traditional human-based platform observations. Possible models include Logistic Regression, Decision Tree, and Random Forest classifiers, with a baseline model used for comparison. I will evaluate the classification models using accuracy, precision, recall, F1-score, and ROC-AUC. In addition, I may use regression models such as Linear Regression, Decision Tree Regression, and Random Forest Regression to predict search interest scores based on time, keyword, category, and platform-related features. These models will not prove direct individual-level replacement, but they can help test whether AI-related and traditional platform trends show distinguishable patterns.

Overall, this project will combine search trend data and survey-based datasets to examine the degree to which people are shifting certain types of interaction from human-to-human to human-to-AI. The final dataset will include both macro-level trend data and large-scale survey data, allowing the project to analyze this shift from multiple perspectives.